UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Strategic Management Accounting

A.Y. 2026/2027

Balanced Scorecard group work

Assignment brief

Silvia Pilonato
Rocco Frondizi

The group work at a glance

3

students in each group

7

cases, one for each group

20

minutes of presentation for each group

0 to 3

points added to the exam mark of each member

Each group analyses one company case and designs a **Balanced Scorecard** for it: a strategy map, objectives, measures, targets and initiatives. It also searches **Scopus** for studies on the scorecard in its sector: the search goes in the slides, but is **not presented** in the 20 minutes.

The work is done autonomously and presented in class; all members present and receive the same score.

Four dates to keep in mind

LECTURE	DEADLINE	LECTURE	LECTURE
10 November	**29 November, 23:59**	**1 December**	**2 December**
Groups and cases are presented	The presentation is sent to the teachers	Discussion: Groups 1, 2 and 3	Discussion: Groups 4, 5, 6 and 7

All members of every group attend **both** discussion lectures, not only the one in which their group presents.

What the work should show

Four perspectives

Learning and growth, internal processes, customer, financial, and the logic that links them.

Value proposition

Operational excellence, customer intimacy or product leadership: which one, and why.

Strategy map

Every arrow is a cause and effect hypothesis that can be tested.

Lag and lead

Outcome measures and the drivers that anticipate them.

Targets and initiatives

Numbers with a date, and actions with an owner.

Sustainability

Where relevant: pick the aspects with double materiality, place each by its path (integration, non-market perspective or derived scorecard), say what sits at the top.

The presentation in five blocks

1	**The company and its strategy**	Mission, vision and strategic choice in your own words; the value proposition; the two or three issues that matter most in the figures.
2	**The strategy map**	Between 10 and 16 objectives, each with at least one cause and effect link. Tell the story from the bottom to the top.
3	**The scorecard**	One table with a row for each objective: measure, calculation, lag or lead, current value, target, initiative, owner.
4	**Implementation**	Who is involved, cascading to at least one unit, reporting frequency, main risk of the design.
5	**The case questions**	Short answers to the four questions at the end of your case.

Rules for the strategy map

Size

Between **10 and 16** strategic objectives across the perspectives.

Objectives specific to this company, not generic.

Links

Every objective has at least **one cause and effect link** to another objective.

Tell the story from learning and growth up to the top.

Special cases

What sits at the top: business case, mission first or balanced. TVO and Valfonda: what replaces profit?

Sustainability: is the aspect relevant? Does it reach the company through markets? Then choose one of the three options, and say why.

The scorecard table: required columns

Objective	Measure	How it is calculated	Lag or lead	Current value	Target	Initiative	Owner
Keep our loyal customers	Repeat customer rate ↑	Customers with two or more purchases in the year ÷ all customers; loyalty card data	Lag	35%	45% by 2028	Loyalty programme for regular customers	Marketing manager

Example of one row, not related to the cases

One or two measures

One per objective, two at most, each with ↑ or ↓.

At least one lead

Every perspective needs at least one lead indicator.

Current value

From the case, or "not measured yet" when the company does not measure it.

Dated targets

Each target has a number and a time horizon.

Literature search on Scopus

Search **Scopus** for studies on how the balanced scorecard is applied in the **sector of your case company**. Read only the **titles and abstracts**: you do not need to read the full papers or comment on their content.

IDEAS FOR THE SECTOR KEYWORDS, TO COMBINE WITH "BALANCED SCORECARD"

1 Torrenti

manufacturing, industrial equipment, servitization

2 Nove Lune

hotel, hospitality, tourism

3 TVO

public transport, bus, public sector

4 Valfonda

dairy, agricultural cooperative, agrifood

5 LPI

logistics, warehouse, third party logistics

6 Ardesia

bicycle, automotive, new product development

7 Contabilis

software, SaaS, cloud, accounting firms

Your own keywords

Starting points only: choose and justify yours.

The search goes **in your slides** (two slides and the PRISMA diagram), after the presentation. It is evaluated, but it is **not presented** and does **not count** in the 20 minutes.

Two slides on your search

SLIDE 1

The method

1. **Keywords used**: and how you combined them, for example "balanced scorecard" AND (hotel OR hospitality), searched in title, abstract and keywords.

2. **Time period**: of the studies considered, for example 2000 to 2026.

3. **Type of sources**: academic articles, reviews, conference papers, professional reports.

4. **Language**: of the sources, for example English only.

SLIDE 2

The results

1. **Total results found**: with your search string and filters.

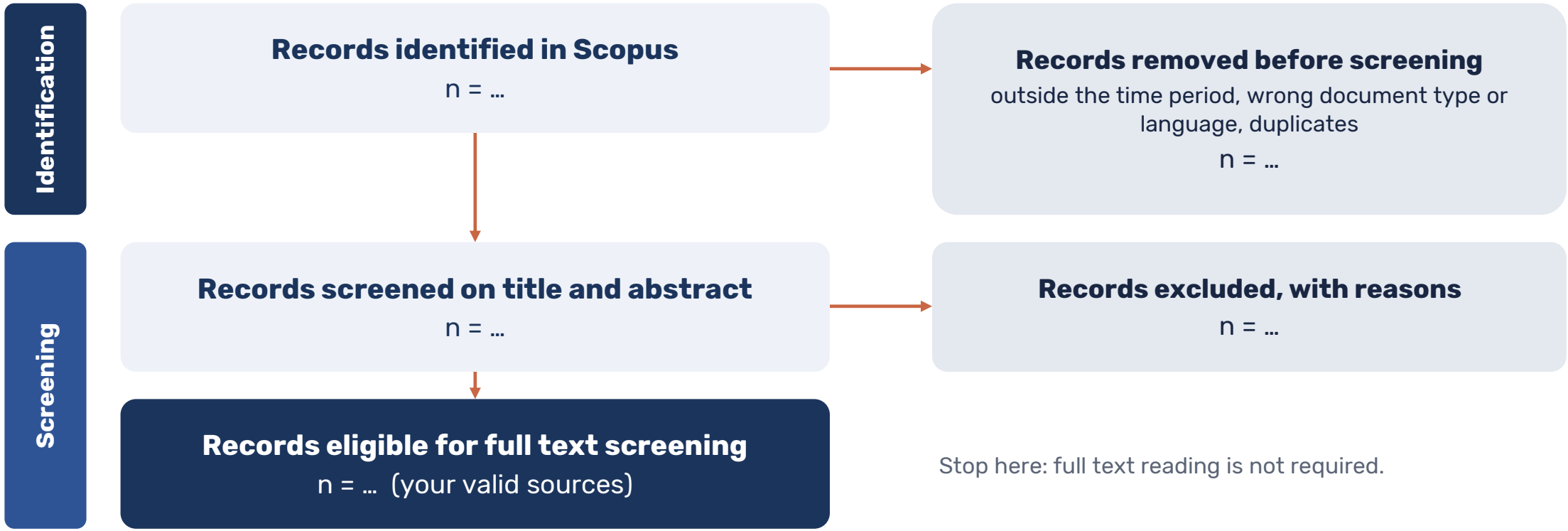
2. **Potentially relevant sources**: and the sources excluded because not relevant to the aims of the search, with the reason.

3. **Valid sources**: their number only, with no comment on content; the full list goes at the end of the presentation.

4. **PRISMA flow diagram**: up to the papers eligible for full text screening.

The PRISMA flow diagram, up to full text screening

Fill in every n. Stop when you have the papers **eligible for full text screening**: these are your valid sources.



Literature search: in the slides, not in the 20 minutes

Presented in class: 20 minutes

In the file only: not presented

Presentation
at most 20 slides

Literature search
2 slides + PRISMA

References
valid sources first

Appendix
if any

IN THE FILE YOU SEND

Everything, in this order

The presentation, then the **two slides on the search** and the **PRISMA diagram**, then the references, with the full list of the valid sources first.

IN CLASS

Only the presentation

The 20 minutes are for the five blocks, from the company to the case questions. Do **not** present the search: it uses **none** of the 20 minutes.

EVALUATION

The search counts

The teachers evaluate the search from the file: method, numbers, PRISMA diagram and reference list.

Format and submission

One PowerPoint file

At most 20 slides (.pptx). After them: the literature search, the references and any appendix. They are **in the file** but **not presented**.

First slide

Group number, case number and company, names and student ID numbers of all members.

Assumptions

One slide called **Assumptions**: write there every number you had to assume.

References

Last slides: the full list of the valid sources from your Scopus search, then every other source used.

File name

Group number and case number, for example **SMA_Group3_Case5.pptx**.

Sending

One member emails both teachers, the others in copy, by **29 November, 23:59**.

The presentation in class

Everybody presents

All members present a part of the work: three in groups 1 to 6, two in group 7.

Time is kept

The presentation stops at **20 minutes**, finished or not.

The literature search is **not presented**: it uses none of the 20 minutes.

Questions follow

Teachers and classmates ask questions after each presentation.

Any member can be asked about any part.

Evaluation: 0 to 3 points

Criterion	What we look at
Strategy	The value proposition and the key issues follow from the facts of the case
Strategy map	Objectives specific to this company; links plausible and explained; sustainability placed where it acts
Measures and targets	Measures precise and with a direction; lead and lag balanced; targets realistic and dated
Implementation and judgement	Cascading and reporting thought through; risks and limits recognised
Literature search	Method clear and replicable; numbers consistent; PRISMA diagram correct; full reference list
Presentation and discussion	Clear slides, time respected, convincing answers from all members

3 points	A coherent scorecard that management could start to use, well defended
2 points	Sound work with some weak links or vague measures
1 points	Basic structure, but generic or not connected to the case
0 points	Not sent on time, or a scorecard that does not use the case

Eight common mistakes

Too many measures

A scorecard with 40 indicators is a list, not a strategy.

Generic objectives

"Improve quality" or "satisfy customers" could belong to any company.

Boxes without arrows

Or arrows that nobody can explain.

Only lag indicators

The scorecard says what happened, not what will happen.

Targets without a date

A target needs a number and a time horizon.

Initiatives that repeat the measure

"Increase customer retention" is not an initiative.

Measures without a direction

A number without ↑ or ↓ is information, not a goal.

Sustainability on the side

Measures listed but not linked to strategy: a symbol, not a tool.

Seven cases, seven strategic problems

Case	Company	Sector	Size
1	Torrenti Valvole S.p.A.	Industrial valves for water utilities and chemical plants	410 employees · €84 million
2	Nove Lune Hotels	Four boutique hotels in Northern Italy	180 employees at peak · €13.6 million
3	Trasporti Valli Orobiche S.p.A.	Public bus transport in a city and its valleys	640 employees · 260 buses
4	Cooperativa Casearia Valfonda	Dairy cooperative making PDO cheeses	140 member farms · €99 million
5	Logistica Padana Integrata S.r.l.	Contract logistics for online retailers and fashion brands	480 employees · €58 million
6	Ardesia Bikes S.p.A.	Premium electric mountain and city bikes	520 employees · €109 million
7	Contabilis Cloud S.r.l.	Cloud software for small accounting firms	120 employees · €18.4 million

The seven companies and all the people in the cases are **fictional**. The figures are realistic but invented for teaching purposes.



UNIVERSITÀ DEGLI STUDI DI BERGAMO | Dipartimento di Scienze Aziendali

Torrenti Valvole S.p.A.

Case 1

Balanced Scorecard group work

Torrenti: the company and its strategy

The company

Founded in 1971, Torrenti makes ball, butterfly and control valves for water utilities, chemical plants and engineering contractors in Italy, Germany and North Africa. For forty years it competed on technical quality and reliable deliveries.

Since 2020 Asian producers have offered standard valves at prices 25% to 30% lower. Torrenti lost two large tenders in 2024 on price alone. The board does not believe it can win a price war.

The new strategy: "From valves to flow"

Launched in 2023 by the CEO, Marta Bonaldi. Torrenti no longer wants to sell a valve and leave: it sells valves fitted with sensors, a remote monitoring platform and multi year maintenance contracts. The customer pays for a network that does not stop.

Targets for 2028: services at **20% of revenue**, at least **150 customers** on a service contract, **ROI back above 12%**.

Torrenti: key figures

	2023	2024	2025
Revenue (€ million)	86.0	84.5	84.0
of which services (€ million)	4.3	5.9	7.6
Gross margin, products	32%	31%	30%
Gross margin, services	46%	44%	45%
Operating income (€ million)	7.7	6.1	5.0
Average invested capital (€ million)	62	64	66
Customers with a service contract	38	61	79
Service contracts renewed at expiry		72%	64%
Orders delivered on time	91%	88%	84%
Warranty claims (% of product sales)	1.1%	1.3%	1.6%
Field service engineers	14	18	19
Turnover of field service engineers	8%	17%	21%
Training hours per field engineer	32	24	18

The minimum required rate of return used by the board is 9%.

Torrenti: what management sees today

The monthly report to the board has four pages: order intake by product line, revenue, gross margin and inventory days. Services are one line, "other revenue". Nobody measures how long a sensor stays offline or how fast an engineer reaches a customer.

Salespeople receive a bonus of 1.5% on the value of the orders they bring in. A five year service contract is worth less in its first year than a single order of valves.

Marta Bonaldi, CEO

“Our customers tell us they love the monitoring. But I cannot see in our numbers whether the plan is working. I only see the margin going down”.

Paolo Rinaldi, sales director

“My people are paid on orders. When a customer hesitates, they give the service contract away free to close the valve order. What else can they do?”

Giulia Sala, head of service

“I lost four engineers last year to a competitor. They drive 40,000 km a year and nobody trains them on the new sensors. At renewal, the customer remembers the day we arrived late”.

Luca Ferri, CFO

“Services grow, but ROI fell from 12% to below 8%. Investors ask me when the plan will pay. I need something better than a promise”.

Your turn

Torrenti: questions for the group

1. Compute ROI and residual income for 2023, 2024 and 2025. What do the two measures tell you, and what do they not tell you, about the new strategy?

2. Which value proposition is Torrenti moving towards? Which measures in the current report support it, and which work against it?

3. Find in the figures a chain of causes and effects from learning and growth to the financial perspective. How could management test whether it really holds?

4. The monthly report shows services as one line, "other revenue". Which measures should replace it to show whether the new strategy works?

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Nove Lune Hotels

Case 2

Balanced Scorecard group work

Nove Lune: the company and its strategy

The company

Started in 2009 by Chiara and Andrea Moroni, who turned a family palazzo in the Città Alta of Bergamo into a hotel of 38 rooms. Today: Bergamo, Verona, Mantova and Lake Como. The Como hotel is seasonal and closes from November to February.

Guests are mostly couples and small groups from Germany, Switzerland, the United States and Italy. Restaurants, a small spa and private tours add about 30% to room revenue.

The strategy: "Every guest is known by name"

Staff remember preferences, suggest a wine bar that is not in the guidebooks, and treat a returning guest as a friend. The promise is a personal stay, not the cheapest room in town.

In 2023 the group hired a revenue manager from an international chain. Prices are now set daily by software, and more rooms are sold through online travel agencies, which charge a commission of **18%** on each booking.

Nove Lune: key figures

	2023	2024	2025
Occupancy	71%	74%	76%
Average daily rate (€)	148	156	162
Revenue per available room (€)	105	115	123
Total revenue (€ million)	11.6	12.8	13.6
Gross operating profit (% of revenue)	33%	31%	29%
Commissions paid to travel agencies (€ million)	1.01	1.17	1.30
Rooms booked directly with the hotel	31%	27%	24%
Guests who had stayed before	22%	19%	17%
Average online review score (out of 10)	8.9	8.7	8.6
Complaints per 1,000 stays	6.1	7.4	8.3
Staff turnover	34%	41%	45%
Staff on seasonal contracts	28%	35%	39%
Guests with a profile in the guest database	18%	21%	23%

Nove Lune: what management sees today

Every morning the four general managers receive occupancy, average daily rate and revenue per available room for the day before. Each month the founders look at gross operating profit by hotel. A general manager earns a bonus of 10% of salary if the hotel reaches its budgeted gross operating profit.

The guest database exists, but the front desk fills in a profile only when there is time. Reviews are read by the marketing assistant and are not discussed in management meetings.

Chiara Moroni, cofounder

“Revenue is up 17% in two years and the bank is happy. So why do I feel we are becoming like any other hotel?”

Stefano Galli, revenue manager

“Rooms are perishable. An empty room tonight is lost forever. The agencies fill rooms we would not sell otherwise, and the average rate keeps rising”.

Elena Conti, general manager in Verona

“I have new people at the front desk every spring. By the time they know the regular guests, the season is over. Nobody pays me to train them; I am paid on profit”.

Andrea Moroni, cofounder

“Our best guests came back every year and told their friends. I want to know how many of them we still have”.

Your turn

Nove Lune: questions for the group

1. Revenue grew by 17% in two years. Is the strategy working? Support your answer with at least four figures from the table.

2. What does the value proposition of Nove Lune require from internal processes and from people? Design the learning and growth perspective for a workforce with 45% turnover.

3. Design the group scorecard and the scorecard of one hotel. Which objectives are the same, and which must change for the seasonal hotel in Como?

4. Is the online review score a lead or a lag indicator? Argue both views and decide where it goes in your map.

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Trasporti Valli Orobiche S.p.A.

Case 3

Balanced Scorecard group work

TVO: the company and its strategy

The company

TVO runs the buses of a mid sized city and of the valleys around it: 58 lines, 2,100 stops, about 23 million passengers a year. All shares belong to public bodies. TVO does not aim to make a profit: its statutes ask for "accessible, reliable and sustainable mobility for the community".

A service contract with the Region fixes 14.3 million vehicle km a year. The Region pays a fixed amount per km and a penalty for every km not operated. Fares cover about one third of the costs.

The challenge

The contract ends in 2027. In 2028 the Region will award the service through a **public tender**, and private operators are expected to bid. The award criteria give 60% of the points to quality (punctuality, reliability, customer satisfaction, emissions) and 40% to price.

TVO cannot find drivers. Young people do not want shifts that start at 5 a.m., and a bus licence costs about €4,000. When drivers are missing, trips are cancelled.

TVO: key figures

	2023	2024	2025
Passengers (million)	21.4	22.8	23.1
Vehicle km operated (million)	14.2	14.0	13.8
Trips cancelled	0.7%	2.1%	3.5%
Buses arriving within 3 minutes of the timetable	86%	83%	81%
Customer satisfaction (out of 10)	6.9	6.7	6.5
Fare revenue (€ million)	18.6	19.5	19.9
Payments from the Region (€ million)	38.5	40.0	41.2
Penalties for km not operated (€ thousand)	120	310	540
Operating costs (€ million)	58.2	61.0	62.4
Operating cost per vehicle km (€)	4.10	4.36	4.52
Drivers employed	412	398	385
Vacant driver positions	18	31	44
Sick leave days per driver	11	13	15
Zero emission buses (out of 260)	9	14	22

Other revenue (advertising and school services) is about €1.7 million a year. The result for the year is close to zero in all three years.

TVO: what management sees today

The board receives a quarterly report prepared for the Region: km operated, penalties, passengers and the budget. Punctuality is measured by the GPS on every bus, but the data are not analysed by line or by time of day. Customer satisfaction is surveyed once a year by an external firm.

In 2024 TVO approved a plan to buy 80 electric buses by 2030, financed in part with public funds. Each electric bus costs about twice as much as a diesel bus.

Roberto Pesenti, general manager

"In 2028 we compete with companies that run buses in ten countries. If we cannot show the Region our quality in numbers, we lose the contract, and the community loses its company".

Anna Locatelli, head of operations

"I plan shifts every week with 44 drivers missing. I usually cancel the evening trips in the valleys, where few people travel. But for those few people, it is the only bus".

Davide Brembilla, HR manager

"We pay the same wage as everybody else. What we could offer is better shifts and help with the licence. Nobody has ever measured what keeps a driver with us".

Sara Gritti, for the municipalities

"For us the question is simple. Does a student in the upper valley reach school on time?"

Your turn

TVO: questions for the group

1. TVO does not aim to make a profit. What sits at the top of its strategy map (business case, mission first or balanced), and where does the financial perspective go?

2. Who is the customer: the passenger, the Region that pays the contract, or the municipalities that own the company? Show how your customer perspective deals with this.

3. Build the chain from the shortage of drivers to penalties and customer satisfaction. Which lead indicators would have warned management a year earlier?

4. Electrification: in the four perspectives or in a non-market one? Emissions target per vehicle km (measures efficiency) or in total (measures real reduction)?

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Cooperativa Casearia Valfonda

Case 4

Balanced Scorecard group work

Valfonda: the company and its strategy

The company

Founded in 1957 by 23 farmers. Today 140 farms bring their milk to two cheese plants, which make Taleggio PDO, other soft cheeses and a mountain cheese aged up to two years, sold in Italy and, more and more, abroad.

The farmers are at the same time the **owners** and the only **suppliers** of milk. The aim is not to maximise profit, but to pay members the best possible milk price, year after year. Whatever remains after costs goes back to members through the milk price.

The challenge

Three retailers in Germany and Switzerland buy 38% of the cheese. In 2025 two of them asked all dairy suppliers to cut emissions per kg by **30% by 2030**, compared with 2025, and to report every year. Suppliers that do not will be dropped.

About 80% of the emissions of a kg of cheese arise on the farms. A biogas plant for the manure of member farms would cost €6 million and cut emissions by about 15%. Meanwhile the number of farms falls every year: many farmers are over 55 and their children do not want to continue.

Valfonda: key figures

	2023	2024	2025
Member farms	152	146	140
Milk collected (million litres)	58	57	55
Cheese produced (tonnes)	5,800	5,700	5,500
Revenue (€ million)	96	101	99
Share of revenue from exports	28%	31%	34%
Price paid to members (€ per 100 litres)	52.0	54.5	53.0
Average price of other dairies in the area (€ per 100 litres)	49.5	51.0	50.5
Farms where nobody will take over the business	31%	34%	37%
Milk from farms certified for animal welfare	22%	35%	41%
Emissions per kg of cheese (kg CO_2e, estimate)	13.8	13.5	13.4
Energy costs of the cheese plants (€ million)	3.1	4.6	3.9

Valfonda: what management sees today

The board, made of nine farmers, meets every month. It looks at sales by customer, the stock of cheese in the aging rooms and the milk price that can be paid next month. Emissions are estimated once a year by the quality manager with a spreadsheet and average values from the literature, not with data from each farm.

Animal welfare certification is voluntary. Certified farms receive €1 more per 100 litres.

Giacomo Arnoldi, president and dairy farmer

"My members ask me one thing at every meeting: the price of milk. If I talk about carbon, they ask who pays for it".

Ilaria Fumagalli, export manager

"In Germany every meeting starts with CO_2. Our cheese is excellent, but a buyer cannot put taste in a sustainability report".

Marco Bianchi, quality and sustainability

"We say emissions went down. In truth our estimate barely moved, and we produce less cheese than before. I am not sure what we are telling the customer".

Federica Rota, young member farmer

"I want to take over my father's farm. But the new barn costs €400,000. I need to know the price of milk for the next ten years, not for next month".

Your turn

Valfonda: questions for the group

1. Valfonda does not aim to maximise profit. What sits at the top of its strategy map (mission first or balanced), and what does its financial perspective measure?

2. Analyse the market and non-market path of the main sustainability issues. On that basis, choose one of the three options for a sustainable balanced scorecard.

3. Compute total emissions in 2023, 2024 and 2025. A target per kg measures efficiency, a total target measures real reduction: which should the retailers ask for, and why?

4. Most emissions arise on member farms, which Valfonda does not manage. How can the scorecard reach the farms? Where does generational renewal belong in your map?

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Logistica Padana Integrata S.r.l.

Case 5

Balanced Scorecard group work

LPI: the company and its strategy

The company

LPI runs the warehouses of 25 clients that sell online: fashion, cosmetics, sports goods and books. It receives and stores the goods, picks and packs every order, ships it and handles returns. Clients pay a price per order, fixed in three year contracts.

Contracts include service levels: at least **99.5%** of orders without errors, and at least **95%** of orders received by 3 p.m. shipped the same day. Below these levels LPI pays penalties.

The strategy

Clients compare prices to the cent. The CEO, Alberto Cattaneo, wants LPI to ship at the lowest cost per order with the fewest errors in Northern Italy: "We will not be the most creative. We will be the most reliable and the cheapest".

In 2025 the board approved **€9 million** for an automatic sorter and picking robots in the second warehouse, to start in 2027. The supplier promises a 20% cut in the cost per order there. The system needs fewer workers, but more technicians.

LPI: key figures

	2023	2024	2025
Orders shipped (million)	6.1	7.4	8.0
Revenue (€ million)	48.0	55.0	58.0
Operating margin	6.1%	4.9%	3.8%
Orders shipped without errors	99.62%	99.55%	99.41%
Orders received by 3 p.m. shipped the same day	97%	95%	92%
Penalties paid to clients (€ thousand)	40	180	420
Clients at year end	22	24	25
Clients lost during the year	1	3	4
Hours worked by agency workers	38%	46%	52%
Injuries per million hours worked	21	26	31
Absenteeism of employees	6.2%	7.0%	7.9%
Technicians able to maintain automated equipment	3	3	4

Correcting a wrong order (return, new shipment, credit note) costs LPI on average €18.

LPI: what management sees today

Each warehouse manager receives a weekly report with orders shipped, hours worked and cost per order. Errors and same day shipments are counted only at month end, when the client reports arrive. Injuries are recorded by the safety officer for the legal register and are not part of the management report.

In November and December volumes triple. LPI then hires up to 300 agency workers for six weeks, with two days of training.

Alberto Cattaneo, CEO

“We earn cents per order. Our only advantage can be to do the same job cheaper and with fewer mistakes than anyone else”.

Francesca Moretti, operations director

“In peak season I fill the floor with people who have never seen a warehouse. Cost per order looks fine in the weekly report, then the errors arrive a month later”.

Nicola Testa, key account manager

“The four clients we lost last year did not leave because of price. They left after Black Friday, when their customers got the wrong parcels”.

Laura Pellegrini, HR and safety manager

“Behind every injury there is a person. Our insurer has also raised the premium by 30%. And next year I need ten technicians I cannot find”.

Your turn

LPI: questions for the group

1. Which value proposition does LPI pursue? Which internal processes are critical for it, and which are less important?

2. Compute the cost of wrong orders in each year (€18 per order) and add the penalties. Classify these costs among the quality cost categories. Which prevention and appraisal activities would you measure?

3. Injuries and agency work: only a social (S) issue in ESG terms, or also a strategic one for LPI? Place them in your map.

4. Which measures would show whether the automation project is on track before its financial results appear? Which measures would change once it runs?

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Ardesia Bikes S.p.A.

Case 6

Balanced Scorecard group work

Ardesia: the company and its strategy

The company

Founded in 2008 by Tommaso Ardesi, a former mountain bike racer, and his partner, an engineer. Ardesia designs its own frames and integrates motor and battery inside the frame, so that an electric bike looks and rides like a normal one. Bikes sell at €3,500 to €9,000 through specialist dealers, who also service them.

During the pandemic Ardesia sold everything it made. Since 2023 the European bike market has been falling: shops are full of unsold bikes and many brands cut prices.

The strategy

Ardesia does not compete on price. It wants to be the **first brand** to bring each new technology to the market: lighter frames, longer battery range, smarter motors. Riders pay more because it is the bike that others will copy a year later.

In 2024 a German competitor launched a lighter integrated frame six months before Ardesia's own model. In 2025 the two new models of the year were both late.

Ardesia: key figures

	2023	2024	2025
Bikes sold	31,000	25,400	22,800
Average selling price (€)	4,580	4,650	4,780
Revenue (€ million)	142.0	118.1	109.0
Operating margin	11.2%	4.5%	1.8%
Research and development (% of revenue)	5.8%	7.2%	8.1%
Engineers	64	78	81
Patents filed	6	9	11
Revenue from models launched in the last two years	46%	38%	29%
Average months from concept to launch	22	26	29
New models launched on schedule	3 of 4	1 of 3	0 of 2
Battery failures in the first year (per 1,000 bikes)	8	14	19
Warranty costs (% of revenue)	1.9%	2.6%	3.4%
Stock at dealers (weeks of sales)	14	23	19
Dealer satisfaction (out of 10)	7.8	7.1	6.8

In 2024 Ardesia changed its supplier of battery cells, saving €22 per battery.

Ardesia: what management sees today

The monthly report shows sales by country and model, margin by model, and the budget of each R&D project. The board reviews R&D through two numbers: total spending and patents filed. Each project has a target launch date, but delays are not reported until the date has passed.

New EU rules on batteries make producers responsible for collecting and recycling used batteries from electric bikes. Ardesia has not yet decided how to organise it with its dealers.

Tommaso Ardesi, founder and CEO

“We have more engineers and more patents than ever. So why are we always second?”

Valentina Ghisleni, R&D director

“My engineers work on three or four projects at the same time. Marketing changes the specifications halfway, and every change costs us months”.

Matteo Ravasio, sales director

“Dealers are full of last year's bikes. To clear them we give discounts, and the brand suffers. A dealer who waits for a new model that never arrives starts selling another brand”.

Silvia Mazzoleni, quality manager

“The new cells saved money on paper. Then the batteries started to fail, and every failure means a dealer, a rider and a replacement battery”.

Your turn

Ardesia: questions for the group

1. R&D spending and patents rise, while revenue from new models falls. What does this say about input and output measures of innovation? Which would you put in the scorecard?

2. Who is the customer of Ardesia: the rider or the dealer? Design a customer perspective that deals with both.

3. Compute warranty costs from 2023 to 2025 and compare them with the savings from the new cells. In which quality cost category do they fall? Which prevention measures belong in your internal process perspective?

4. Should the collection of used batteries enter the scorecard now? Is it relevant, does it reach Ardesia through markets, and which of the three options fits?

UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Contabilis Cloud S.r.l.

Case 7

Balanced Scorecard group work

Contabilis: the company and its strategy

The company

Founded in 2012 in Brescia by Federica Bonetti, an accountant, and a software engineer. Contabilis sells cloud software to small accounting firms: bookkeeping, payroll, tax returns and electronic invoicing. Firms pay a monthly subscription for each user; there is no licence to buy.

When electronic invoicing became compulsory between Italian businesses in 2019, many small firms moved to the cloud and Contabilis grew fast. Today it serves 4,300 firms with 120 employees.

The strategy: “The accountant’s best colleague”

Contabilis is not the cheapest software. It promises two things: every new tax rule is in the software **before the deadline**, and a support team of former accountants who answer quickly, also in the busiest weeks of the year.

Large software groups now offer bundles at lower prices, and several are adding artificial intelligence features. Contabilis wants to keep its customers by being closer to them than anyone else.

Contabilis: key figures

	2023	2024	2025
Customers at the start of the year	2,200	2,900	3,600
New customers during the year	870	961	1,168
Customers lost during the year	170	261	468
Annual recurring revenue at year end (€ million)	11.8	14.9	17.6
Total revenue (€ million)	15.2	17.0	18.4
Operating margin	14.0%	9.1%	4.2%
Sales and marketing spending per new customer (€)	1,100	1,450	1,900
Support requests per customer in the year	2.2	2.6	3.3
Average time to the first answer (hours)	4	9	17
Software releases with a critical error	1 of 10	3 of 11	6 of 12
Developers at year end	38	41	40
Developers who left during the year	9%	17%	24%
Training hours per support agent	30	18	12
Net promoter score (from −100 to +100)	41	29	18

On average a customer pays about €4,100 a year.

Contabilis: what management sees today

Every month the founders look at new customers, recurring revenue and costs. The support team is measured by the number of requests closed by each agent every day; the developers by the number of new features released.

Nobody analyses why customers leave: a firm that cancels fills in a form with one free text field, which nobody reads. The satisfaction survey is sent once a year, in September.

Federica Bonetti, founder and CEO

“We sign more firms than ever. So why does the margin fall every year?”

Luca Locatelli, head of customer support

“In the two weeks before a tax deadline requests triple. My agents close them fast, but the same customer calls back three times”.

Davide Rota, chief technology officer

“Every month we release new features to keep up with the large groups, and we have no time to fix what we released last month. My best developers leave for the banks”.

Elena Carminati, sales director

“New customers come easily: our name is good. But it is hard to sell to a firm whose colleague next door has just left us”.

Your turn

Contabilis: questions for the group

1. Customers and recurring revenue grow, while margin and net promoter score fall. Compute, for each year, the share of customers lost, the recurring revenue lost with them (€4,100 each) and the spending to acquire new customers. Is the strategy working?

2. Which value proposition does Contabilis pursue? Which internal processes are critical for it, and which of the current measures (requests closed per agent, features released) work against it?

3. Build the chain from developers leaving and training hours to errors, answer times, lost customers and recurring revenue. Which lead indicators would have warned management in 2024?

4. Data security and the workload of the team before tax deadlines: only ESG issues, or strategic for Contabilis? Is each relevant, does it reach the company through markets, and which of the three options fits?